

DESIGN AND ANALYSIS OF ALGORITHMS

Practical No. 5: Kruskal's Minimum Spanning Tree

1. Aim

To find the Minimum Spanning Tree (MST) of a connected weighted graph using Kruskal's algorithm.

2. Objective

- Understand the greedy approach for graph problems.
- Construct an MST using minimum-weight edges.
- Avoid cycles using a disjoint-set technique.

3. Theory

Kruskal's algorithm is a greedy algorithm for finding a Minimum Spanning Tree of a connected, undirected, weighted graph. All edges are sorted by non-decreasing weight. The smallest edge is selected if it does not form a cycle with already selected edges. The process continues until $V - 1$ edges are selected. With efficient sorting and disjoint-set operations, the complexity is commonly $O(E \log E)$.

4. Conclusion

Kruskal's algorithm constructs an MST by making locally optimal edge selections while preventing cycles. It demonstrates the effectiveness of the greedy strategy in network optimization.